

ISO/IEC 17025 standards

International Organisation for Standardisation (ISO) issued ISO/IEC 17025 as the primary standard used for testing and calibration laboratories. These standards specify the general requirements for the competence to carry out tests and calibrations using standard methods, non-standard methods, and laboratory-developed methods. There are many commonalities between ISO 17025 and ISO 9000 standards, but the latter does not include the competence to the equation.

All the accredited laboratories around the world follow these ISO 17025 standards to improve their ability to consistently produce valid results. ISO 17025 includes the technical requirements that determine the correctness and reliability of the tests and calibrations performed in a laboratory.

the date of previous valid calibration," shares Chandmal Goliya, managing director, Kusam Mecco. He recommends recalibration at the earliest to avoid further errors if the readings in between are suspect.

Citing the case of how low-end, low-cost equipment are handled, Goliya says, "The user tends to handle these equipment with very less care and precaution. As these are already low-budget equipment, such mishandling disturbs the accuracy and needs frequent recalibration. Mishandling results in erroneous results apart from damaging these equipment."

Some equipment have a highly sensitive electronic circuit. For example, a digital multimeter may be calibrated once in six months, while an automated test equipment or a tester may need full calibration of all its boards and resources once in two weeks. In equipment that have time-base oscillators involving frequency counters and sensors, signal generators, spectrum analysers, precision digital multimeters, network analysers and LCR meters, drifts are caused very easily by usage and environmental conditions. Such equipment need calibration at least once a year to meet their specifications and accuracy.

Desai feels, "The user has to fix the periodicity after seeing the results for three to four consecutive calibrations. For example, if every year it is exactly the same or within limits, he can extend the periodicity. If every year it becomes worse and he has to adjust or write down the correction chart, he has to compress the schedule. Gener-

ally, electronic equipment periodicity is one year, while for mechanical items like micrometers it is six months. Very precision equipment may require three months. If the equipment undergoes any kind of repair, it has to be calibrated after repair."

"Any equipment which has to sustain shocks, vibrations, violent change of temperature or pressure, gets disturbed. Hence the master is kept in the lab in an ideal environment. Masters that are used for on-site calibration and are subjected to vibrations and change in environment due to travelling or handling, are checked and corrected with masters in the labs after they come back to base," explains Desai.

"Normally, manufacturers of electronic instruments specify accuracy for a period of time. It is advisable that a user maintains and follows the calibration schedule and gets instruments calibrated at regular intervals recommended by the manufacturer," cites Kela.

Advancements in microprocessor and microcontroller technology, sensor technology along with an intelligent software algorithm and connectivity to computers have led to the development of a new feature called 'self-calibration' and 'auto-tuning.'

"Today, test and measurement equipment are available with inbuilt calibration check points that can be used as reference points for getting a calibration done. Calibration in itself is the next big opportunity in the Indian test and measurements industry as there are very limited options available for the industry," cites Manish Kwatra,

CEO, MetroQ Instruments.

In-house calibration vs third-party labs

It's an ongoing debate in many manufacturing as well as R&D organisations whether to have an internal calibration lab or outsource the calibration activity to an external accredited lab. There are situations where you may require to re-calibrate the equipment every day. It is totally subjective to the nature of organisation and use of measurement equipment.

"Whether you manage internally or externally, boils down to a business assessment. Large shops are likely to save money by outsourcing, while small installations may find it less expensive to manage it internally," comments Keith Kidd, director-packet systems testing at Vizor Labs.

Calibration is a process and needs to be managed like any other activity in the organisation. Kidd adds, "Regardless of how one manages the calibration process, it is necessary to ensure that your training and controls for the calibration system are well known to all. The most likely hiccup with calibration is getting the organisation to follow the process, which takes discipline."

Philip Wright, director at QSO, a weighing equipment and scale service company, explains a case of his customer for whom his company carried out scale and weighing equipment calibration twice a year. The organisation had its own test weights which were also calibrated on a yearly basis.

"We used to carry out the equipment calibrations and issue a certificate after completion of test once or twice a year. However, on a daily basis, at the start of every shift the operator using the equipment would place the required amount of the test weights onto the equipment as a check and, if necessary, re-calibrate the equipment himself," he tells. You may be thinking what was the point in having an outside company come in and calibrate the equipment in the first place then.